Fulcrum

Technology Due Diligence Report

Prepared for Illustrative Purposes

Target: Meridian Software Ltd
Sector: B2B SaaS — Logistics & Supply Chain
Revenue (LTM): £18.2M
Employees: 142
Prepared for: [PE Fund Name]
Date: April 2026
Assessment period: 10 working days

> ILLUSTRATIVE SAMPLE — This report uses a fictional company to demonstrate the Fulcrum assessment methodology, deliverable format, and depth of analysis. All data is fabricated for demonstration purposes.

1. Executive Summary

Meridian Software Ltd operates a B2B SaaS platform for logistics route optimisation and warehouse management, serving 340+ enterprise clients across the UK and DACH region. The platform processes ~2.1M daily API calls and manages £4.2B in annual freight value for its clients.

OVERALL ASSESSMENT

Architecture	4/5
Team & Organisation	3/5
Security & Compliance	2/5
Data & Analytics	3/5
Cost Structure	4/5
Delivery Capability	4/5
Strategic Alignment	3/5
Composite	**3.3/5**

TOP 3 RISKS

1. Security posture: SOC 2 non-compliant, 3 critical CVEs unpatched
No SOC 2 Type II. Penetration test (Nov 2025) identified 3 critical and 8 high-severity vulnerabilities. IAM relies on shared service accounts for production database access. Estimated remediation: £285K over 6 months.

2. Key-person dependency: 2 engineers hold 68% of core routing engine knowledge
The route optimisation algorithm (Meridian's core IP) was built by two senior engineers who have been with the company since founding. No documentation. No tests on critical paths. Bus factor = 2. Mitigation requires £120K investment in documentation, pair programming, and knowledge transfer over 4 months.

3. Architecture scalability ceiling at ~3.8x current load
The monolithic API handles 2.1M daily calls on vertically-scaled AWS instances. Horizontal scaling is not supported without architectural refactoring. At the value creation plan's projected growth rate, the ceiling is reached in Month 18. Refactoring estimate: £440K over 12 months.

TOP 3 VALUE CREATION OPPORTUNITIES

1. Cloud cost optimisation: £210K annual savings (currently 2.3x industry benchmark)
AWS spend is £68K/month (£816K/year). Industry benchmark for this revenue band: £25K–30K/month. Key drivers: oversized RDS instances, no reserved capacity, 40% of S3 storage is orphaned data. 60-day implementation.

2. AI-powered route optimisation: £2.1M incremental ARR opportunity
Current routing algorithm is rule-based (2019 vintage). ML-based routing using existing data could improve accuracy by 12–18%, enabling premium tier pricing. Data quality is sufficient. Estimated build: £380K over 9 months. Payback: 7 months.

3. Data monetisation: benchmark dataset across 340+ logistics operators
Meridian processes routing and cost data from 340+ logistics companies. This dataset has intrinsic value as an industry benchmark product (fuel cost trends, route efficiency comparisons, seasonal patterns). Estimated development: £180K. Potential ARR: £600K–1.2M within 24 months.

RECOMMENDATION

RECOMMENDATION: GO WITH CONDITIONS

Technology supports the investment thesis at current scale. Three material conditions: (1) security remediation must begin within 30 days of close (£285K), (2) key-person risk mitigation plan required as Day 1 priority, (3) architecture refactoring must be budgeted in the value creation plan (£440K, Month 3–15). Total cost-to-fix: £1.84M over 24 months. Recommended price adjustment: £1.2M (net of cloud savings in Year 1).

COST-TO-FIX SUMMARY

PHASE	INITIATIVES	INVESTMENT	TIMELINE	EXPECTED ROI
P1: Quick wins	Cloud cost reset, security patching, IAM hardening	£295K	0–3 months	3.8x (Year 1)
P2: Foundation	SOC 2 programme, knowledge transfer, data governance	£665K	3–12 months	2.4x
P3: Growth enablement	Architecture refactoring, ML routing, benchmark product	£880K	12–24 months	4.1x
Total	**9 initiatives**	**£1.84M**	**24 months**	**3.2x blended**

2. Company Technology Profile

FIELD	DETAIL
Company	Meridian Software Ltd
Sector	B2B SaaS — Logistics & Supply Chain
Founded	2017 (Birmingham, UK)
Revenue (LTM)	£18.2M (92% recurring)
ARR growth	34% YoY
Employees	142 total (38 engineering, 12 product, 92 commercial/ops)
Clients	340+ enterprise accounts, 94% UK/DACH
Daily API calls	~2.1M
Core product	Route optimisation + warehouse management SaaS
Cloud provider	AWS (eu-west-2, London region)
Annual tech spend	£4.1M (22.5% of revenue)

TECHNOLOGY STACK

LAYER	SYSTEMS	NOTES
Frontend	React 18, TypeScript, Next.js	Modern, well-maintained
Backend	Python (Django), Node.js microservices	Monolith (Django) + 4 microservices (Node)
Database	PostgreSQL (RDS), Redis, Elasticsearch	RDS oversized (see Cost section)
Infrastructure	AWS EC2, RDS, S3, CloudFront, SQS	No IaC. Manual deployments to production
CI/CD	GitHub Actions	Automated for staging, manual for prod
Monitoring	Datadog (basic tier)	APM not configured. Alerting gaps
CRM	HubSpot	Well-integrated
Analytics	Metabase (self-hosted)	Limited adoption. Most reporting is manual

3. Architecture & Infrastructure

Score: 4/5 (Good) — Modern stack with solid fundamentals. Key concern: monolithic API limits horizontal scaling.

CURRENT ARCHITECTURE

Meridian runs a hybrid architecture: a Django monolith handling the core routing engine and client API, plus four Node.js microservices (notifications, billing, data export, webhook relay). All services run on AWS EC2 in eu-west-2 (London). The frontend is a React/Next.js application served via CloudFront.

TECHNICAL DEBT INVENTORY

ITEM	SEVERITY	REMEDIATION COST	TIMELINE
Monolith API: no horizontal scaling	High	£440K	12 months
No Infrastructure as Code	High	£60K	2 months
Manual production deployments	Medium	£35K	1 month
Python 3.9 (EOL Dec 2025)	Medium	£25K	3 weeks
Orphaned S3 buckets (2.4TB)	Low	£5K	1 week
Total technical debt		**£565K**	

SCALABILITY ASSESSMENT

Current architecture handles 2.1M daily API calls on two vertically-scaled EC2 instances (r5.4xlarge). Load testing indicates the ceiling at approximately 8M daily calls (~3.8x current). At the value creation plan's projected 40% annual growth, this ceiling is reached in approximately Month 18. Horizontal scaling requires extracting the routing engine from the Django monolith into a dedicated service with stateless request handling.

SINGLE POINTS OF FAILURE

SPOF	IMPACT IF FAILURE	CURRENT MITIGATION
Primary RDS instance	Full platform outage	Multi-AZ failover (automated, tested quarterly)
Routing engine EC2	API unavailable	Auto-scaling group (2 instances), but no cross-AZ
Redis cache	Degraded performance (10x latency)	ElastiCache single-node. No replication

4. Team & Organisation

Score: 3/5 (Moderate) — Capable team with critical key-person risk. Engineering management layer is thin.

ENGINEERING COMPOSITION

ROLE	HEADCOUNT	IN-HOUSE	OUTSOURCED	NOTES
Backend	14	10	4	Outsourced via Polish agency (2-year relationship)
Frontend	8	8	0	Strong React team
DevOps / Infra	3	3	0	Understaffed for platform complexity
Data / ML	2	2	0	Recent hires (< 6 months tenure)
QA	4	2	2	Manual testing predominant
Engineering management	3	3	0	VP Eng + 2 team leads. No dedicated architect
Product	4	4	0	CPO + 3 PMs
Total	**38**	**32**	**6**	**84% in-house**

KEY-PERSON RISK ASSESSMENT

PERSON	ROLE	RISK	RATIONALE
James Chen	Principal Engineer (routing)	Critical	Co-architect of routing engine. Only person who understands the constraint solver. No documentation.
Sarah Kovacs	Senior Engineer (routing)	Critical	Second routing expert. Together with Chen, holds 68% of core IP knowledge. No succession plan.
Marcus Webb	VP Engineering	High	All architecture decisions route through Marcus. No deputy. Hiring authority bottleneck.

Mitigation recommendation: Immediate documentation sprint (4 weeks, £40K). Pair programming programme between Chen/Kovacs and two mid-level engineers (ongoing). Recruit a dedicated Solutions Architect to reduce VP Eng bottleneck (£80K/year).

5. Security & Compliance

Score: 2/5 (Significant) — Material gaps across compliance, vulnerability management, and access controls. This is the primary risk dimension.

COMPLIANCE STATUS

STANDARD	STATUS	GAP	REMEDIATION COST
SOC 2 Type II	**Not started**	Full programme required. Estimated 6–9 month timeline to certification.	£180K
GDPR	**Partial**	Privacy policy compliant. Data processing agreements missing for 3 sub-processors. No DPIA completed.	£25K
ISO 27001	Not started	Not required for current client base but increasingly requested in enterprise RFPs.	£120K

VULNERABILITY ASSESSMENT

External penetration test conducted November 2025 by NCC Group. Key findings:

SEVERITY	COUNT	STATUS	EXAMPLES
Critical	3	Unpatched	SQL injection in legacy reporting endpoint, exposed AWS credentials in client-facing error messages, unauthenticated admin API endpoint
High	8	3 patched, 5 open	Weak session management, missing CSRF tokens, outdated TLS configuration
Medium	12	7 patched, 5 open	Information disclosure, verbose error messages, missing security headers

IDENTITY & ACCESS MANAGEMENT

- **Production database:** 4 shared service accounts with full read/write access. No individual credentials. No audit logging on database queries.
- **AWS console:** 8 IAM users, 3 with admin privileges. MFA enabled for 5 of 8 users.
- **Application:** Role-based access control implemented. SSO via Google Workspace. Adequate.

Immediate action required: Patch 3 critical CVEs within 7 days of close. Revoke shared database credentials and implement individual access with audit logging within 30 days. Enable MFA for all AWS IAM users immediately.

6. Cost Structure

Score: 4/5 (Good) — Total tech spend is within range, but cloud costs are 2.3x industry benchmark. Significant optimisation opportunity.

ANNUAL TECHNOLOGY SPEND

CATEGORY	ANNUAL (GBP)	% OF TOTAL	BENCHMARK	ASSESSMENT
Engineering salaries	£2,280K	55.6%	50–60%	Within range
Cloud infrastructure (AWS)	£816K	19.9%	8–12%	**2.3x benchmark**
SaaS licences	£340K	8.3%	6–10%	Within range
Outsourced development	£420K	10.2%	5–15%	Within range
Other (hardware, travel, training)	£244K	6.0%	3–8%	Within range
Total	**£4,100K**	**100%**		**22.5% of revenue**

Tech spend as percentage of revenue (22.5%) is slightly above the 18–22% benchmark for B2B SaaS companies at this revenue level (Gartner, 2025). The variance is driven entirely by cloud infrastructure overspend.

CLOUD COST OPTIMISATION

INITIATIVE	CURRENT ANNUAL	OPTIMISED	SAVINGS	EFFORT
RDS right-sizing (r5.4xl → r5.2xl)	£186K	£93K	£93K	1 day
Reserved instances (1-year, all-upfront)	£420K	£294K	£126K	1 week
S3 lifecycle + orphan cleanup (2.4TB)	£38K	£12K	£26K	2 days
Dev/staging environment scheduling	£96K	£38K	£58K	1 week
Total cloud savings	**£740K**	**£437K**	**£303K**	**~3 weeks**

£303K annual savings achievable within 60 days. No engineering risk. No product impact. This is the single highest-ROI quick win in this assessment.

7. Value Creation Roadmap

90-DAY PLAN (POST-CLOSE)

WEEK	ACTION	OWNER	INVESTMENT	EXPECTED OUTCOME
1	Patch 3 critical CVEs	DevOps	£5K	Eliminate immediate breach risk
1–2	IAM hardening: MFA, revoke shared credentials	DevOps	£10K	Audit-ready access controls
2–4	Cloud cost reset: RDS right-sizing, reserved instances	DevOps	£15K	£219K annual savings (RDS + RI)
3–6	S3 cleanup + dev environment scheduling	DevOps	£8K	£84K additional annual savings
4–8	Documentation sprint: routing engine	Engineering	£40K	Key-person risk from Critical to High
6–12	SOC 2 Type II programme launch	External + VP Eng	£180K	Certification in Month 9–12

12-MONTH TECHNOLOGY PLAN

QUARTER	FOCUS	INVESTMENT	EXPECTED IMPACT
Q1	Security remediation + cloud optimisation + documentation	£258K	£303K annual savings, security posture 2/5 → 3/5
Q2	SOC 2 programme + IaC migration + Python upgrade	£275K	Enterprise sales unblocked, deployment time –80%
Q3	Architecture refactoring (Phase 1) + ML routing PoC	£320K	Scalability ceiling raised to ~10x, ML routing validated
Q4	ML routing production launch + benchmark product design	£280K	Premium tier launch (£2.1M ARR opportunity)

Appendix: Methodology

ASSESSMENT FRAMEWORK

This report follows the Fulcrum 7-Dimension Technology Due Diligence Framework. Each dimension is assessed against standardised criteria and scored on a 1–5 scale:

SCORE	LABEL	DEFINITION
1	Critical	Immediate risk to business operations. Emergency intervention required.
2	Significant	Major gaps impacting value creation plan. 6-month remediation.
3	Moderate	Functional but suboptimal. 12-month improvement roadmap.
4	Good	Minor gaps. Optimisation opportunities exist.
5	Excellent	Best-in-class for company size and stage.

DATA SOURCES

- Source code review (GitHub repository, 14 repositories)
- AWS console access (read-only, 3 days)
- CI/CD pipeline review (GitHub Actions)
- Datadog monitoring dashboards
- Architecture documentation (Confluence, partial)
- Organisation chart and team data (HR system export)
- Financial data: technology spend breakdown (CFO-provided)
- NCC Group penetration test report (November 2025)
- Incident logs: PagerDuty (12 months)
- Product roadmap (Linear)
- Stakeholder interviews: CEO, VP Engineering, CPO, 2 team leads, DevOps lead (6 sessions, 8 hours total)

LIMITATIONS

This assessment was conducted over 10 working days based on information provided by Meridian's management team and direct observation of systems. Fulcrum did not conduct independent penetration testing (relying on the NCC Group report) or audit financial statements. Findings reflect conditions as of the assessment date.

DISCLAIMER

This report is prepared solely for the use of [PE Fund Name] in connection with the proposed acquisition of Meridian Software Ltd. The findings and recommendations represent Fulcrum's professional judgment as of the report date. This report should not be relied upon by any other party without Fulcrum's prior written consent.

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